

Secure Environment for Academic Assessments



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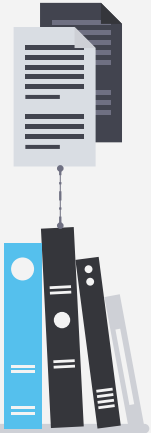
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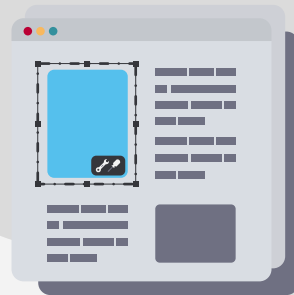
Demonstration





01

State of the Art





Online Proctors

Positive Takeouts

- Continuous Screen Monitoring;
- Personalized Individual Evaluation;
- Allows Remote Exam.

Negative Takeouts

- Individual Human Supervising;
- Paid License;
- Software Based Solution.





Transforming Exams

Positive Takeouts

- Prevents Access to Personal Directories;
- Prevents Access to Online Resources;
- No Prior Software Installation;
- Allows Offline Backup.

Negative Takeouts

- Relies on Student's Personal; Computers Resources;
- Discontinued Solution;
- Paid License.

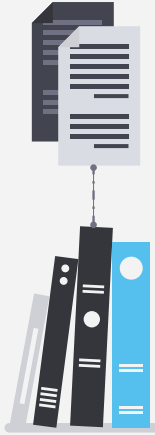
Safe Exam Browser

Positive Takeouts

- Easy Installation;
- Supports Any OS;
- Free-to-Use software.

Negative Takeouts

- Software Based Solution;
- Known Vulnerabilities;
- Relies on a LMS.



Exam Control



Positive Takeouts

- Current Best Solution;
- Files Monitoring;
- Good Security Practices.

Negative Takeouts

- Relies on Institution's Infrastructure;
- Needs previous preparation.





EEL – Electronic Exam Lab

Positive Takeouts

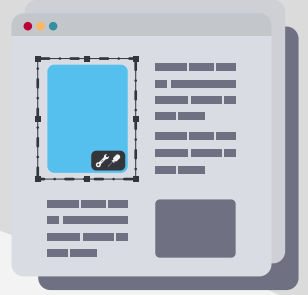
- Good System Concept with a Functional Interface to enhance the user experience;
- Real Time Monitoring.

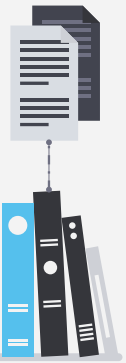
Negative Takeouts

- System is Unable to Perform its Concept;
- Relies on Student's Personal Computers Resources.

02

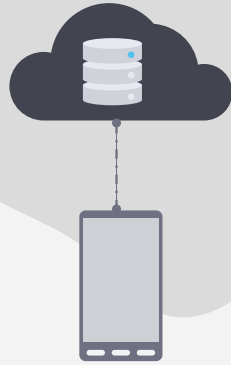
System Scope





Project Scope

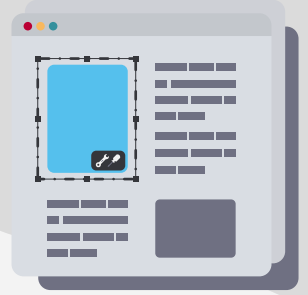
- Focuses only on in-classroom use in a **controlled application content**;
- The Devices are provided by the respective university teaching team and collected at the end of each use;
- Non-presential activities, such remote access or extracurricular use are outside the scope of the project;
- This approach **minimizes vulnerabilities associated with the use of devices in less controlled contexts**, aligning with the principles of digital security and protection of data. With this, it is ensured that the system promotes a teaching-learning experience to the educational environment.





03

Requirements & Use Cases



Functional Requirements

Web App

- Create exams in an intuitive and easy way;
- Be able to upload the exam, any documentation needed and define the time and duration;
- Monitor if all students are well connected to provide any help if necessary;
- Monitor student's progress and activity in real-time to ensure all rules are being followed;
- Users authentication and identification.



Functional Requirements

Isolated System

- Should just get access to what is proposed by the professor and start the exam;
- Make backups every certain amount of time;
- Connect and disconnect easily so if an error occurs don't damage the personal device;
- Ensure all Personal Data will not be accessed by the exam environment thereby preventing any malware or harmful software.



Functional Requirements

Server

- Block Every Connection with Exterior;
- Transfer Data Between Isolated System and WebApp;
- Authorize Student Access to Sessions.



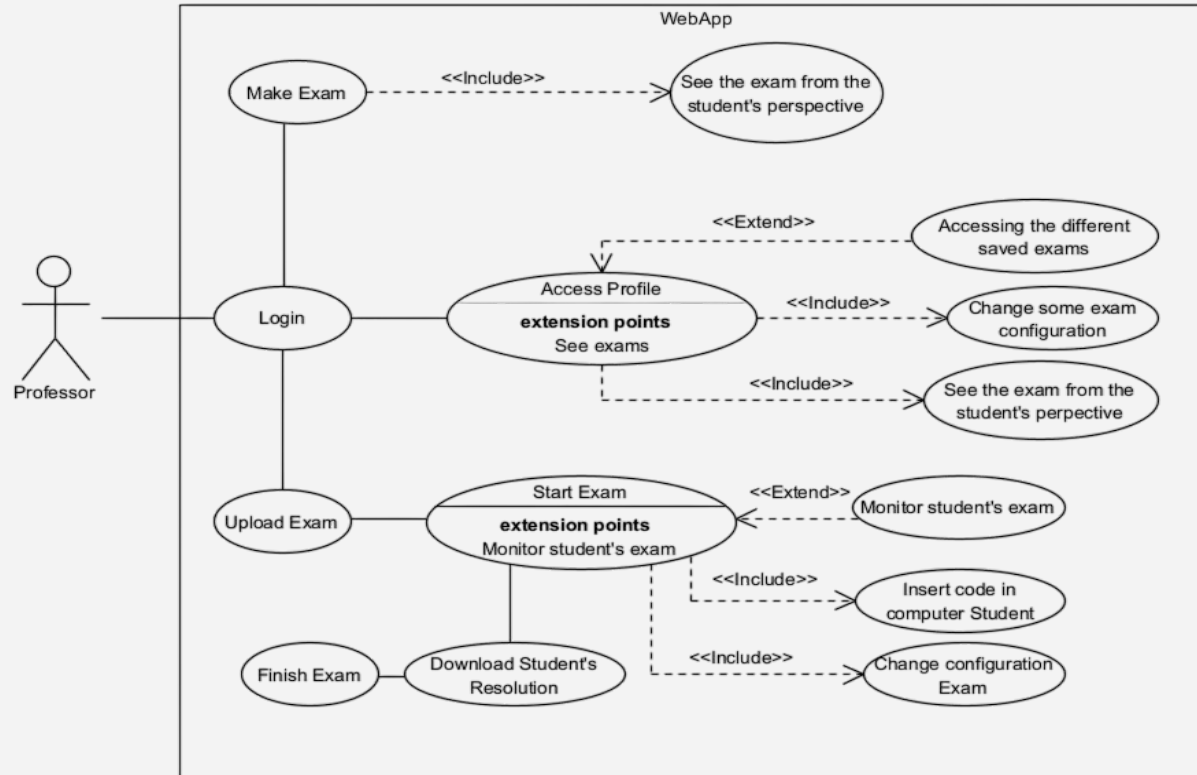
Non-Functional Requirements

- Usage of UA idp for authentication;
- Secure communication and data storage;
- Network and Power failures tolerance;
- Ensure Compliance with GDPR to avoid legal issues;
- Being able to maintenance the system without disturbing exam session.



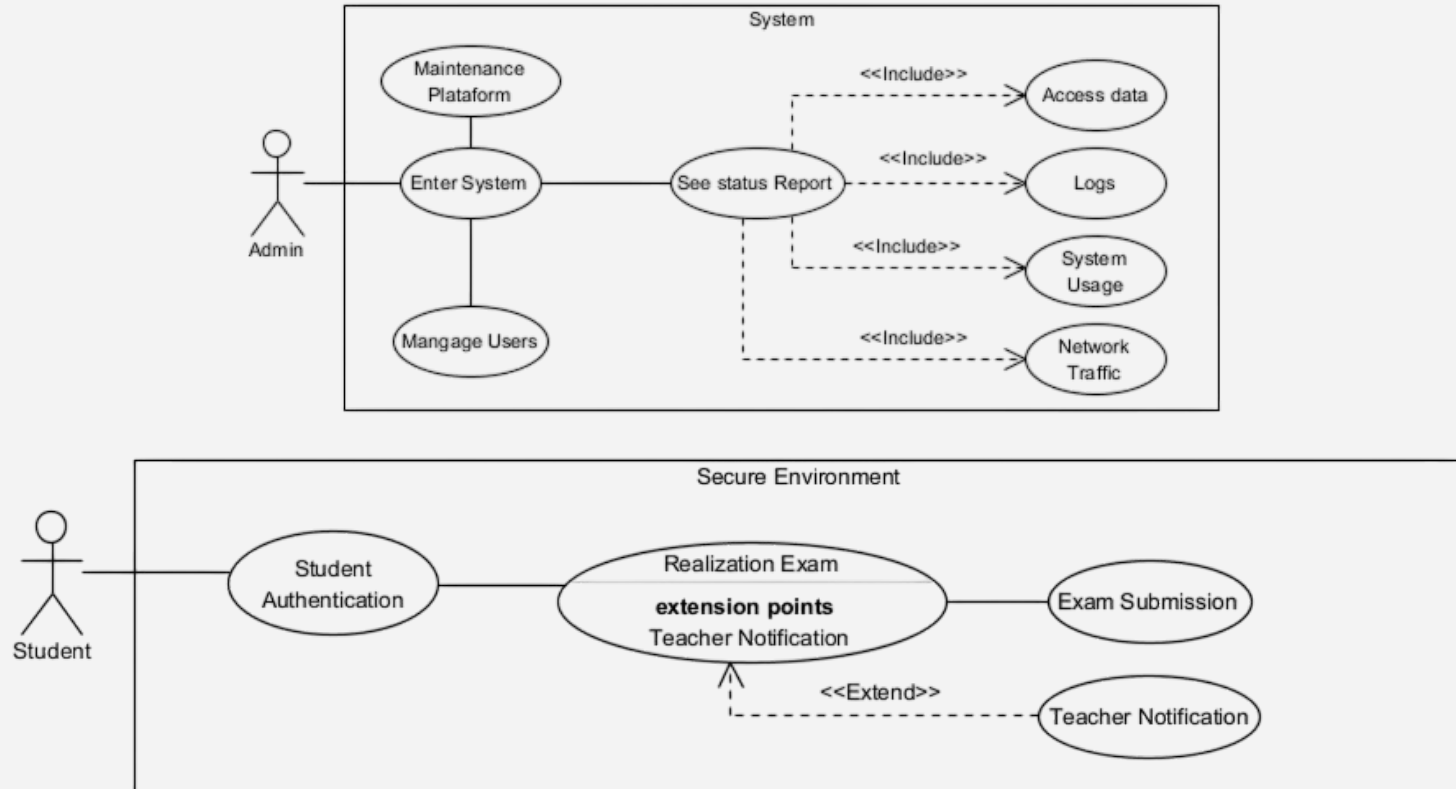


Use Cases



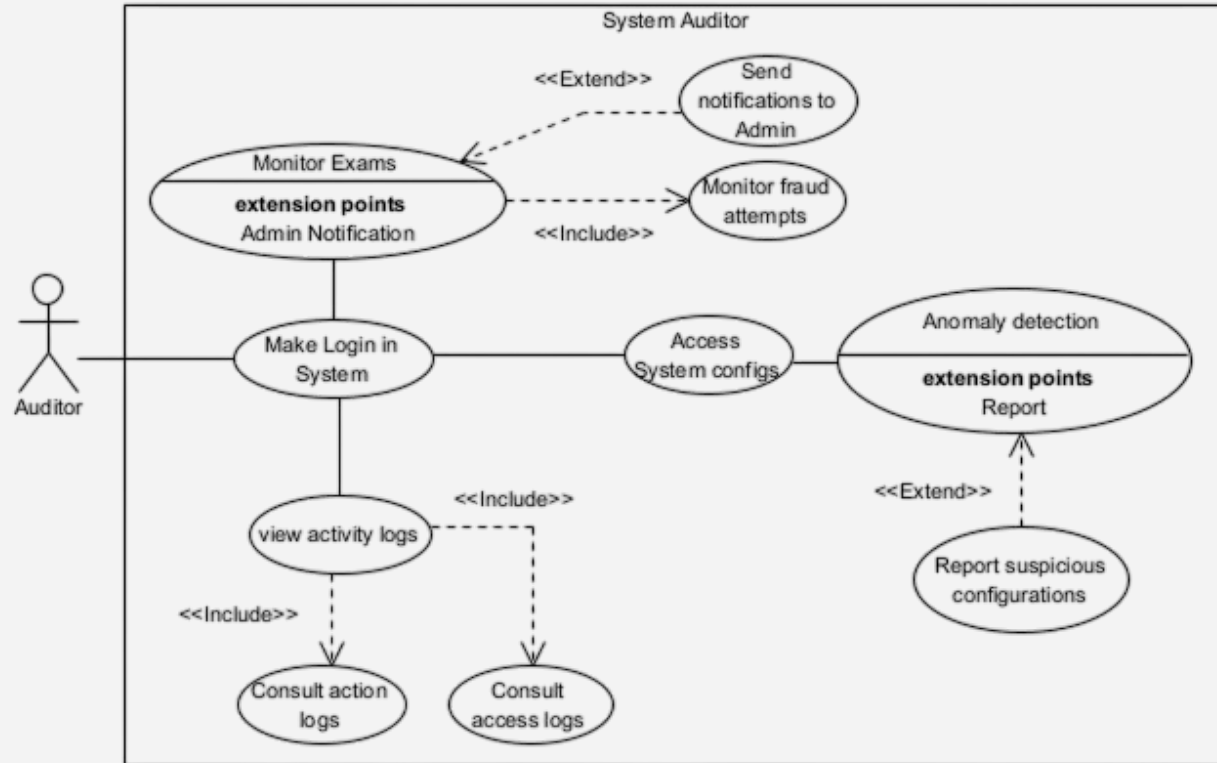


Use Cases



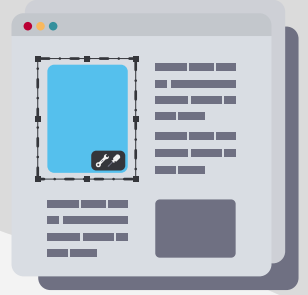


Use Cases



04

Abuse Cases



Abuse Cases



Student:

- Submission of plagiarized or automatically generated assignments;
- Exploitation of system vulnerabilities to gain unfair advantages.

Abuse Cases



Professor:

- Exploitation of direct configuration mode access to install unauthorized software;
- Manipulation of data to obscure performance or inefficiencies;
- Unauthorized retrieval of confidential student information.



Abuse Cases

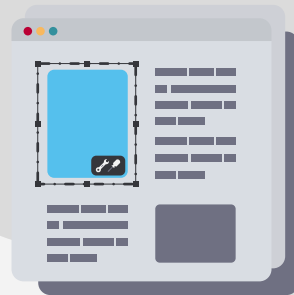
Administrator:

- Alteration of permissions or deletion of sensitive data;
- Creation of fictitious accounts to manipulate security reports.



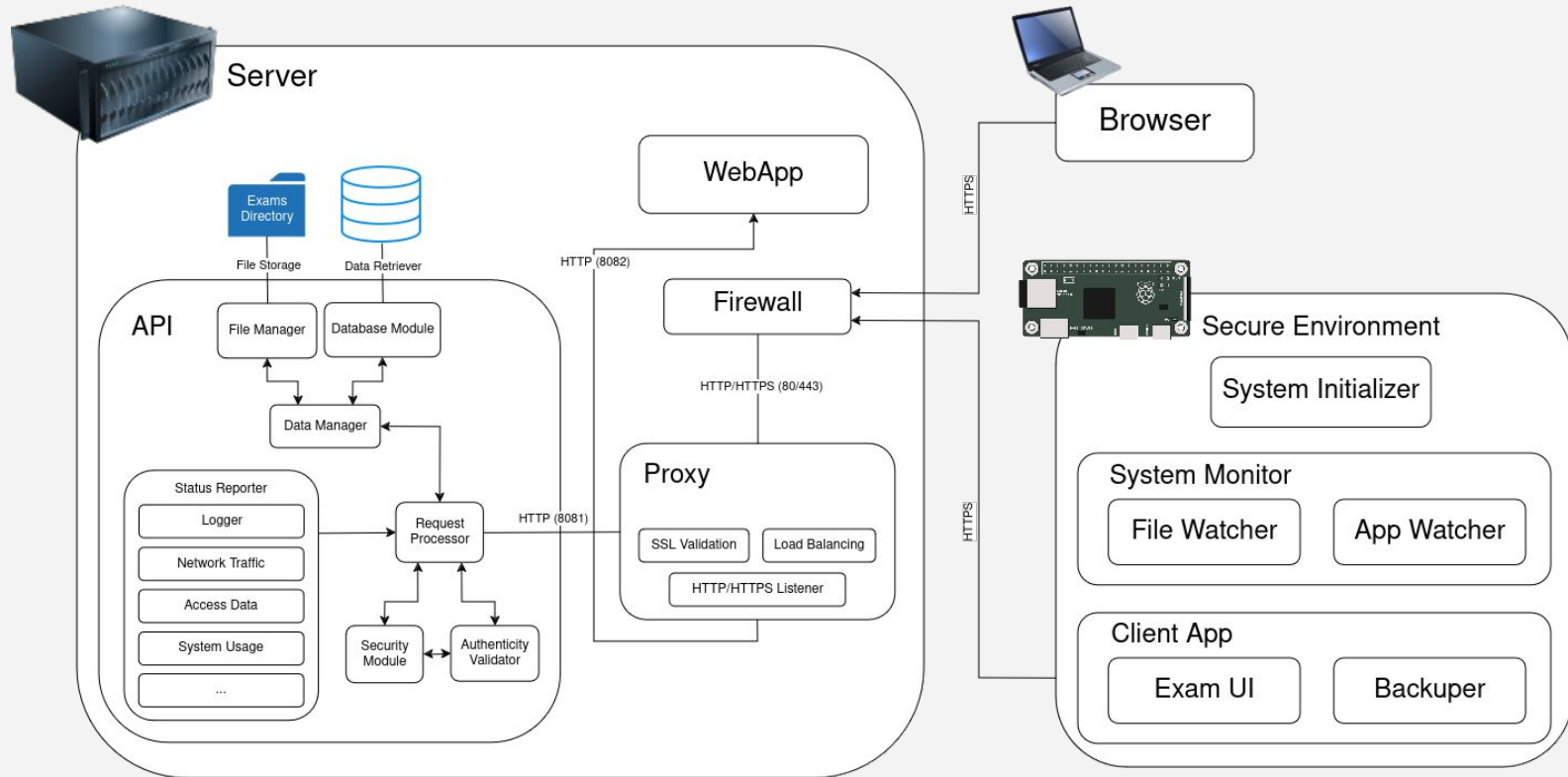
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Architecture



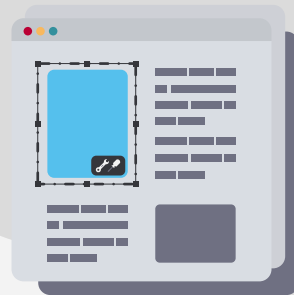


Architecture



06

Demonstration



THANKS!

Do you have any questions?



<https://portableisolatedenvironment.github.io/>

